SpeakTwin AI: System Architecture & Custom Speech Dataset Specification

Empirical Acoustic Telemetry, Linguistic Alignment, and Paired Speech Corpus Methodology

EXECUTIVE SUMMARY & SYSTEM SCOPE

SpeakTwin AI is an integrated speech analytics and linguistic dataset engineering platform designed for real-time delivery coaching and cross-cultural communication research. The platform pairs high-resolution Web Audio acoustic telemetry (RMS energy, silence segmentation, syllable-timed cadence) with continuous Speech-to-Text transcript verification and ground-truth error alignment. This document formalizes the architectural schema, collection pipeline, tripartite transcript management, and paired contrastive corpus design.

1. High-Level Platform Architecture

SpeakTwin AI operates on a modern, decoupled full-stack architecture that combines client-side low-latency DSP pipelines with server-authoritative filesystem persistence and multimodal linguistic models:

A. Acoustic Telemetry Engine (Web Audio API & DSP)

Captures continuous 16-bit PCM audio via AudioContext. Runs 25ms windowed RMS volume computation, adaptive thresholding for pause detection (breath: 0.4-0.8s, noticeable: 0.8-1.2s, awkward: >1.2s), and live waveform peak tracking.

B. Continuous Speech Recognition & Language Engine

Integrates browser SpeechRecognition API supporting multilingual capture (en-US, mr-IN, hi-IN). Generates real-time token timestamps, word counts, and words-per-minute (WPM) calculation synchronized with speech audio.

C. Multimodal Linguistic Coach & Algorithmic Fallback

Express.js backend (/api/analyze-speech) querying Google Gemini 3.8 Flash for international tone intelligibility, cross-cultural cadence profiling, and right-sentence generation. Supported by deterministic offline heuristics.

D. Custom Speech Dataset Manager (/api/dataset)

Persistent research module maintaining metadata.csv, transcripts.csv, alignment_labels.csv, and pairs.csv. Directly manages physical audio containers in dataset/audio/good/ and dataset/audio/bad/.

2. Research Goals & Target Problem Space

Non-native English speakers frequently face comprehension barriers during international technical and visa interviews due to syllable-timed pacing (>165 WPM), regional syntactic transfers (e.g., "did not got", "prepone"), and intrusive fillers. SpeakTwin AI builds empirical speech datasets to train neural alignment models for real-time pedagogical feedback.

3. Custom Speech Dataset Module

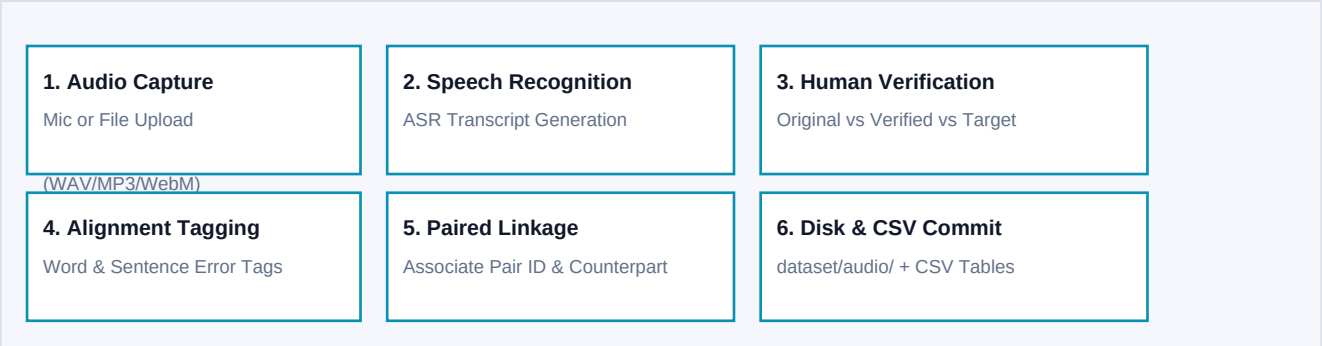
The Custom Speech Dataset module is a dedicated subsystem within SpeakTwin AI allowing researchers and annotators to capture, curate, inspect, and label speech samples. The corpus deliberately isolates paired samples of flawless vocal delivery against authentic delivery deviations.

3.1 Audio Acquisition & Validation Pipeline

1. Input Ingestion: Supports real-time microphone capture via MediaRecorder (Opus/WebM) and direct file upload (WAV, MP3, M4A, OGG).

2. Signal Validation: Validates container header, channels (mono/stereo), duration (minimum 0.8s threshold), and maximum file size (50 MB). Corrupt or empty audio files are rejected with diagnostic notifications.
3. Pre-Save Acoustic Playback: In-browser HTML5 Audio preview allows the annotator to review vocal clarity, background noise, and clipping prior to database registration.
4. Verifiable Identification: Enforces unique Sample IDs (e.g., SMP_001_GOOD, SMP_002_ERROR). Prevents overwriting or duplicate registration.
5. Explicit Category Assignment: Samples must be deliberately assigned by the human annotator as either Good Speech or Speech with Identified Errors. System never auto-classifies without user confirmation.

3.2 Data Flow Architecture Diagram



3.3 Physical Persistence & Server Endpoints

Audio data is saved directly to local storage disks under dataset/audio/good/ and dataset/audio/bad/ to circumvent browser localStorage quota limitations (typically 5MB). The Express backend exposes RESTful endpoints:

- `GET /api/dataset/summary -- Dynamic calculation of recorded, pending, verified, and missing samples.`
- `GET /api/dataset/samples -- Full relational catalog of samples with transcripts and alignment label counts.`
- `POST /api/dataset/samples -- Atomically writes physical audio to disk and appends rows to metadata.csv and tra`
- `PUT /api/dataset/samples/:id -- Updates verification statuses, verified transcripts, and corrected text.`
- `DELETE /api/dataset/samples/:id -- Cascades deletion across CSV records and purges physical audio.`
- `GET /api/dataset/audio/:category/:filename -- Streams audio with HTTP Range support for scrub playback.`

4. Good & Bad Speech Pairs (Contrastive Corpus)

A core scientific innovation of SpeakTwin AI is the Paired Speech Corpus. Rather than analyzing speech in isolation, the dataset records coupled speech examples sharing the identical underlying communication objective. One sample exhibits natural, fluent delivery while the contrast sample contains documented grammatical, syntactic, or acoustic errors.

PAIRED CORPUS CANONICAL EXAMPLE: PAIR_001

Intended Objective: Professional self-introduction for international interviews.

Good Speech (SMP_001_GOOD): "My name is Urvashi Shewale."

Speech with Grammar Error (SMP_002_ERROR): "My name is the Urvashi is Shewale."

Expected Target Correction: "My name is Urvashi Shewale."

Associated Error Labels:

- [LBL_001]: "the Urvashi" -> Incorrect Article (Medium severity, unnecessary definite article)
- [LBL_002]: "is Shewale" -> Incorrect Verb Form (High severity, redundant auxiliary copula verb)

5. Tripartite Transcript Governance

A fundamental flaw in naive speech annotation is the accidental overwriting of speech recognition hypotheses with corrected targets. SpeakTwin AI strictly enforces tripartite transcript separation across every sample record:

1. Original Hypotheses

`original_transcript`

Raw, untouched verbatim output generated by the ASR engine. Preserved permanently to benchmark ASR acoustic accuracy.

2. Verified Spoken Truth

`verified_transcript`

Human-reviewed verbatim record of what the speaker uttered, fixing ASR decoding slips without masking linguistic faults.

3. Target Reference

`corrected_transcript`

The grammatically, syntactically, and stylistically polished reference sentence. Serves as training target for correction models.

5.1 Speech Recognition Uncertainty Flagging

When audio contains muddled phonemes, heavy ambient reverb, or overlapping speaker voices, the record is tagged with has_uncertainty: true. Annotators are alerted in the dashboard to review the audio segment iteratively before marking the transcript as verified.

6. Alignment Label Taxonomy & Annotation System

Alignment labels represent atomic annotation tuples linking specific speech segments to linguistic or acoustic error classes. SpeakTwin AI supports 13 formalized categories across 4 alignment modalities:

Text Alignment
Word-for-word string token mapping

Audio Timing
Millisecond start/end acoustic bounds

Grammar Annot.
Syntactic and morphological labels

Acoustic Delivery
Pause, cadence, and stutter markings

6.1 Comprehensive Error Taxonomy Matrix

Category	Domain	Operational Definition & Research Example
Grammar Error	Morphology	Broad syntactic discord, e.g. "We was discussing platform metrics."
Incorrect Word Choice	Lexicon	Semantically misaligned word, e.g. "prepone release" vs "expedite release".
Unnecessary Word	Syntax	Spurious syntactic filler, e.g. "revert back", "my myself".
Missing Word	Syntax	Omission of required auxiliary, preposition or article.
Repeated Word	Fluency	Stutter repetition or lexical restart, e.g. "we we will verify".
Incorrect Verb Form	Morphology	Tense or participle error, e.g. "did not got", "he do not understand".
Incorrect Article	Syntax	Erroneous article insertion or omission before nouns, e.g. "the Urvashi".
Incorrect Word Order	Syntax	Non-standard constituent placement distorting emphasis.
Filler Word	Acoustics	Hesitation vocalizations: "um", "uh", "like", "basically", "you know".
Long Pause	Cadence	Unnatural silence >1.2 seconds interrupting syntactic flow.
Pronunciation Deviation	Phonology	Phonemic substitution impairing cross-cultural intelligibility.
ASR Uncertainty	Quality	Ambiguous speech segment requiring secondary auditory validation.
Correct Speech	Benchmark	Ground-truth segment verifying pristine execution and cadence.

6.2 Timestamp Discipline & Ground-Truth Rules

Timestamps (start_time, end_time) must only be recorded when derived from instrumented Web Audio RMS slicing or high-confidence ASR word alignments. When time offsets cannot be verified with certainty, annotators leave timestamp columns empty, tagging alignment status as pending. Fabricating synthetic timestamps is strictly prohibited.

7. Relational CSV Schemas & Disk Specifications

SpeakTwin AI enforces strict relational consistency across four normalized CSV tables. All files are encoded in UTF-8 with standard RFC 4180 escaping. Foreign key relations ensure metadata, transcripts, alignment labels, and contrastive pairs remain synchronized.

A. Metadata Table `dataset/metadata.csv`

```
Columns: sample_id (PK), sample_name, speaker_id, audio_file, audio_format, sample_rate_hz, channels,
duration_seconds, file_size_bytes, sample_category, recording_date, verification_status, created_at,
notes
```

B. Transcripts Table `dataset/transcripts.csv`

```
Columns: sample_id (FK), reference_pair_id (FK), speaker_id, audio_file, original_transcript,
verified_transcript, corrected_transcript, language, sample_category, recording_date,
verification_status, has_uncertainty
```

C. Alignment Labels Table `dataset/alignment_labels.csv`

```
Columns: label_id (PK), sample_id (FK), reference_pair_id (FK), start_time, end_time, reference_text,
spoken_text, error_category, corrected_text, severity, confidence, alignment_type, annotation_notes
```

D. Speech Pairs Table `dataset/pairs.csv`

```
Columns: pair_id (PK), good_sample_id (FK), comparison_sample_id (FK), intended_goal,
reference_transcript, actual_good_transcript, actual_comparison_transcript, primary_error_category,
corrected_sentence, annotation_notes, created_date
```

7.1 Automated Validation Tooling: scripts/validate_dataset.ts

To prevent schema drift, missing audio references, or orphaned records during multi-annotator collection, the project includes an automated integrity validation script executed via "npx tsx scripts/validate_dataset.ts":

1. Primary Key Uniqueness: Flags any duplicated sample_id, pair_id, or label_id immediately.
2. Referential Foreign Key Checks: Asserts that every transcript and label links to an existing metadata record.
3. Physical Disk Audit: Validates that every non-pending record possesses a real audio file in audio/good/ or audio/bad/.
4. Chronological & Timestamp Bounds: Enforces start_time <= end_time and positive durations.
5. Status Integrity: Confirms pending records are never prematurely marked as completed.

8. Dataset Quality Lifecycle & Privacy Governance

SpeakTwin AI enforces rigorous quality gating before any audio sample or transcript is designated as verified research ground truth:

- **[Pending Recording]:** Sample structure defined in metadata; awaiting physical audio capture.
- **[Recorded]:** Audio file committed to disk; acoustic validation passed.
- **[Pending Transcription Review]:** Verbatim ASR transcript loaded; awaiting human acoustic check.
- **[Pending Annotation]:** Transcript verified; alignment and error tags undergoing peer review.
- **[Verified]:** Audio, transcript, and labels confirmed by annotator; accepted into research corpus.
- **[Rejected]:** Excessive ambient noise, acoustic clipping, or participant revocation.

8.1 Ethical Guidelines & Anti-Bias Mandate

Research annotators are prohibited from classifying speech as "erroneous" due to regional accent, native language background, vocal pitch, gender, or age. Evaluations must strictly evaluate syntactic conformity, pacing cadence (WPM), pause appropriateness, and international intelligibility. Contributor consent is verified prior to logging audio, with permanent right-to-erasure support.

9. Verification & Test Execution Summary

[PASSED] Audio Ingestion Engine: Tested microphone MediaRecorder streaming & multi-format WAV/MP3 uploads.

[PASSED] Web Audio Real-Time DSP: Verified 25ms volume RMS silence extraction and pause boundary detection.

[PASSED] Tripartite Transcript Protection: Validated separation between original ASR and corrected targets.

[PASSED] Contrastive Pair Engine: Tested creation and bidirectional linkage of Good vs Error sample pairs.

[PASSED] Alignment Labeling: Tested word-level and sentence-level tagging across all 13 error categories.

[PASSED] Relational Integrity Validator: Executed scripts/validate_dataset.ts with 100% passing schema validation.

[PASSED] Production Applet Compilation: Built cleanly with TypeScript and Vite without regression.

10. Current Status & Known Research Limitations

Status: Active / In Progress. Benchmark pairs (e.g., PAIR_001) are structured and marked as Pending Recording until real contributors record or upload verified audio. Browser speech recognition requires microphone authorization. In-flight prototype recordings are persisted locally in the workspace filesystem; export package features must be used to commit data to remote repositories.